MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INTERFACE BUS, KEY CODE PHP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic interface bus module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 811I, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.743 failures per million operating hours for class I or 7.930 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be PHP.

3.5.2 Size. The module size shall be 2A, format B.

3.5.3 Weight. The module shall have a maximum weight of 85 grams.

3.5.4 Contact pins. The module shall have 100 contact pins positioned 2A1 through 2A100.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/341-1 for class I and M28787/341-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

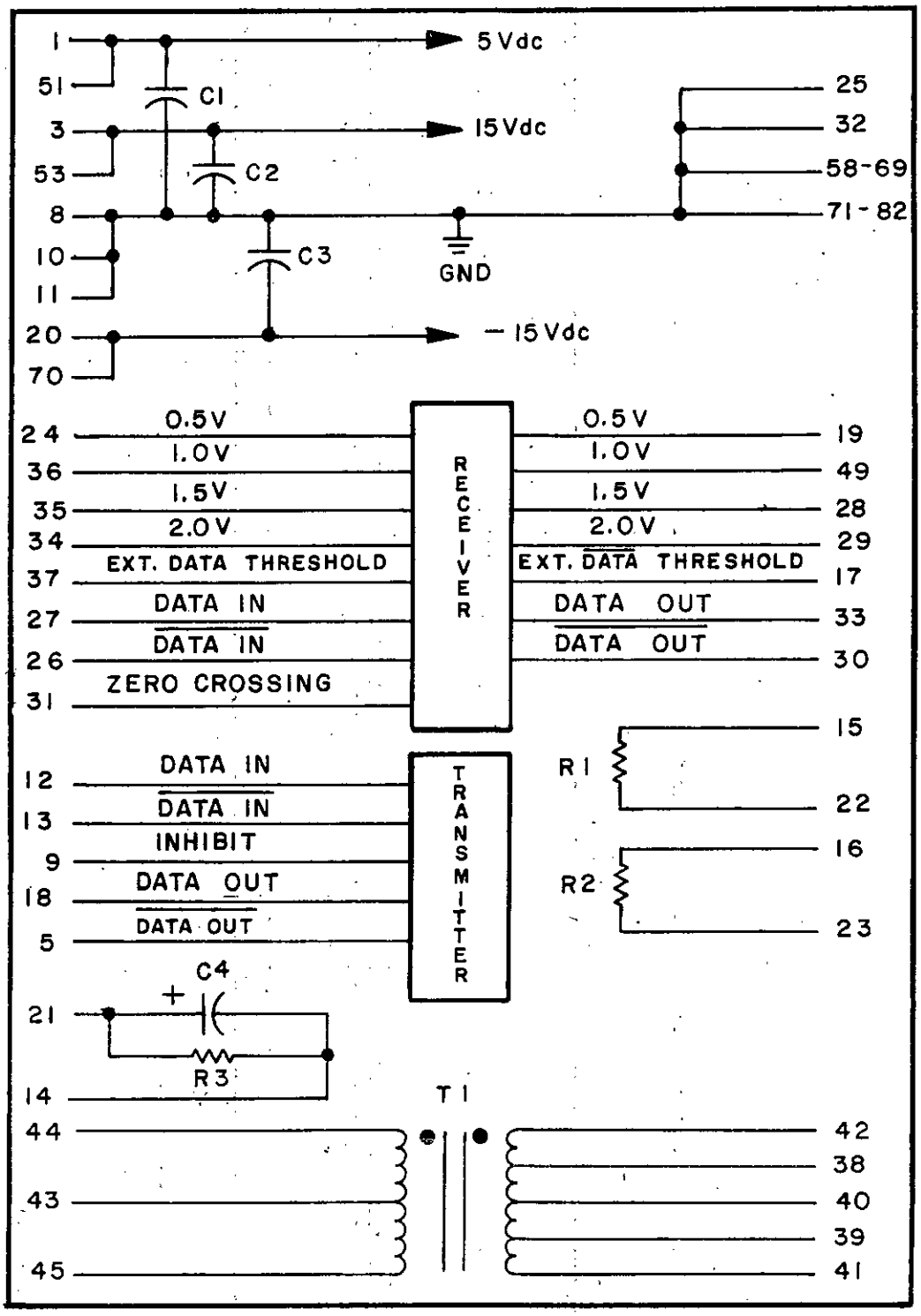
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTE: Pins 2, 4, 6, 7, 46-48, 50, 52, 54-57, and 83-100 are not used.

FIGURE 1. PHP module function diagram, interface bus.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A13	Isolation													
	Initial	II	X	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	II	X	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B31	Continuity	III	X	0	1	0	1	0	1	0	1	0	1	Ω
C1	Power supply current (5 V)	IV	X	1	72	1	72	1	72	1	72	1	72	mA
C2,C3	Power supply current (±15 V)	IV	X	1	95	1	95	1	95	1	95	1	95	mA
D1,D2	Resistance													
	Initial	V	X	54.35	55.45	54.1	55.7	54.1	55.7	53.9	55.9	53.9	55.9	Ω
	End-of-life	V	X	54.1	55.7	53.8	56.0	53.8	56.0	53.6	56.2	53.6	56.2	Ω
E	Impedance (RC network) check													
	Initial	VI	X	2.4	3.8	2.3	4.0	2.3	4.0	2.2	4.5	2.2	4.5	V rms
	End-of-life	VI	X	2.3	4.0	2.2	4.5	2.2	4.5	2.1	4.9	2.1	4.9	V rms
F1-F4	Impedance (transformer) check	VII	X	6.5	13.0	6.5	13.0	6.5	13.0	6.5	13.0	6.5	13.0	V rms
F5,F6	Output voltage (transformer)	VII	X	12.87	13.13	12.87	13.13	12.87	13.13	12.87	13.13	12.87	13.13	V rms
G1,G5, G25	Receiver functional test	VIII	X	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
G2,G6, G26	Receiver functional test	VIII	X	0	800	0	800	0	800	0	800	0	800	mV
G3,G4, G7-G24	Receiver functional test	VIII	X	0	300	0	300	0	300	0	300	0	300	ns
H1	Transmitter functional test	IX	X	0.9	1.1	0.9	1.1	0.9	1.1	0.9	1.1	0.9	1.1	µs
H2,H3	Transmitter functional test	IX	X	100	300	100	300	100	300	100	300	100	300	ns
H4,H5	Transmitter functional test	IX	X	0	750	0	750	0	750	0	750	0	750	ns
H6,H7	Transmitter functional test	IX	X	0	300	0	300	0	300	0	300	0	300	mV

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
H8	Transmitter functional test	IX	X	3.0	4.5	3.0	4.5	3.0	4.5	3.0	4.5	3.0	4.5	V pk
H9	Transmitter functional test	IX	X	-3.0	-4.5	-3.0	-4.5	-3.0	-4.5	-3.0	-4.5	-3.0	-4.5	V pk
J1,J5, J11	Receiver common mode test	X	S	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	2.4	5.0	V
J2,J6, J12	Receiver common mode test	X	S	0	800	0	800	0	800	0	800	0	800	mV
J3,J4, J7-J10	Receiver common mode test	X	S	0	300	0	300	0	300	0	300	0	300	ns
K1,K2	Power supply filter capacitance													
	Initial	XI	X	5.4	8.2	5.1	8.5	5.1	8.5	4.8	8.8	4.8	8.0	µF
	End-of-life	XI	X	5.1	8.5	4.8	8.8	4.4	8.8	4.4	9.2	4.4	9.2	µF
K3	Power supply filter capacitance													
	Initial	XI	X	12	18	11.2	18.7	11.2	18.7	10.5	19.5	10.5	19.5	µF
	End-of-life	XI	X	11.2	18.7	10.5	19.5	10.5	19.5	9.7	20.2	9.7	20.2	µF
L1-L12	Output symmetry	XII	Q	0	90	0	90	0	90	0	90	0	90	mV pk

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through L shall be performed in any order with the following exception. If test condition A is part of a sequence of tests, the order of testing shall be A, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through H and K shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through K shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through L shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation			
Stand by	3.2	3.2	W
Transmitting	6	6	W
Receiving	3.2	3.2	W
Input voltage			
Low level	800 max	800 max	mV
High level	2 min	2 min	V
Input current (V_{IL} = 0.4 V)			
Data in and $\overline{\text{Data in}}$	700 max	700 max	µA
Inhibit	360 max	360 max	µA
Output voltage (line to line)			
Direct coupled mode	6 min, 9 max	6 min, 9 max	Vp-p
Transformer coupled	18 min, 27 max	18 min, 27 max	Vp-p
Propagation delay time			
Transmitter	750 max	750 max	ns
Receiver response	300 max	300 max	ns

6.5 Module function. This module contains a Manchestor coded data transmitter and receiver with an isolation transformer for receiving and transmitting information over a MIL-STD-1553 data bus.

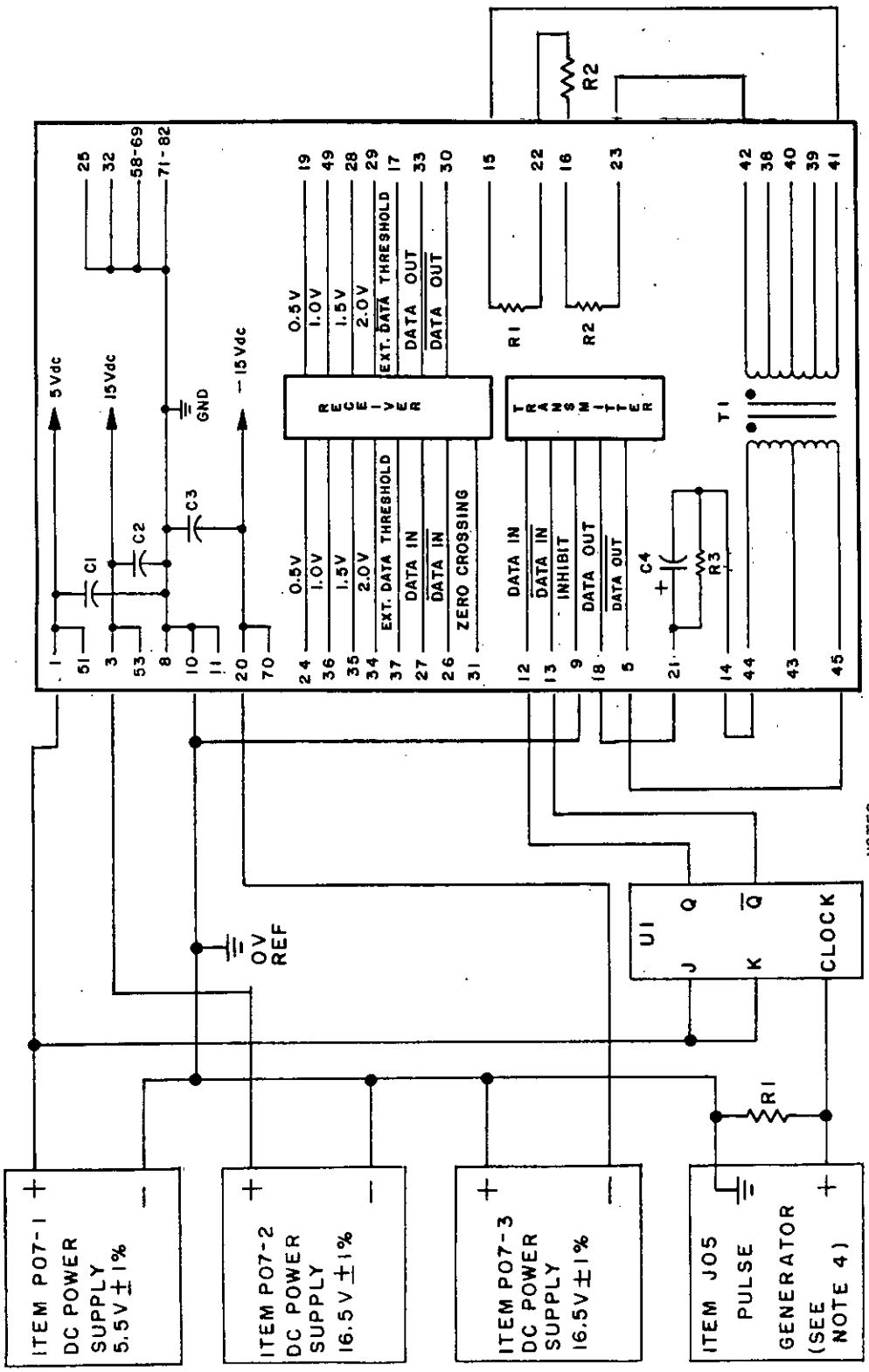
6.5.1 Input to output relationship. The receiver accepts a Manchestor encoded signal from a MIL-STD-1553 data bus and translates it into TTL serial data for processing by a bit processor module PHQ. The transmitter translates serial TTL data from the PHQ module into Manchestor encoded signals and transmits this data on the MIL-STD-1553 bus. The module operates in either direct or transformer coupled mode.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N390)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 35Ω ±2%, 1/8 W.
3. U1 = SN7470 or equivalent
4. Pulse characteristics:
   PW = 625 ±60 ns.
   PRP = 1.25 ±0.12 µs (800 kHz).
   Transition times ≤ 20 ns.
   PA = 4.5 ±0.5 V.
   Baseline = 0.0 +0.4, -0 V.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

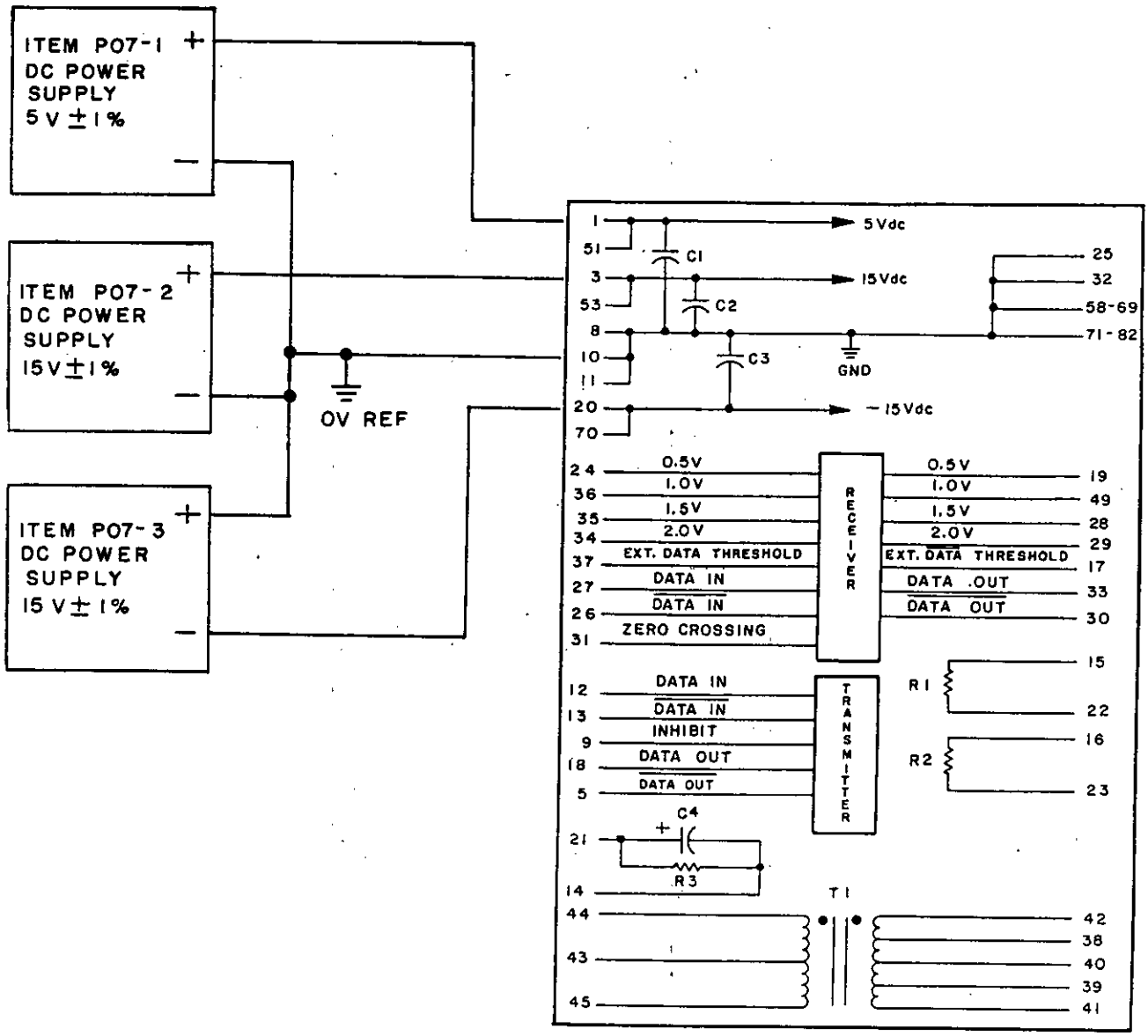


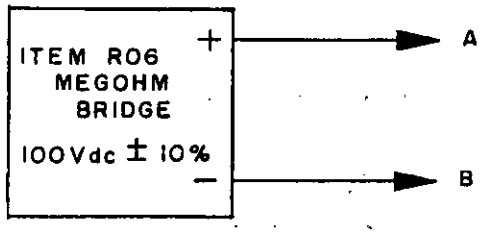
FIGURE 3. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	2,4,6,7,46-48,50, 52,54-57,83-100	1,3,5,8,9-45,49,51,53, 58-82	10	∞	MΩ	4
A2		38-45	1-37,46-100				
A3			14,21				
A4			15,22				
A5			16,23				
A6			Frame				
A7		2	4,6,7,46-48,50,52, 54-57,83-100				
A8		47	46,48				
A9		83	84-100				
A10		54	55-57				
A11		6	7				
A12		4	54				
A13		50	100				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test lead.

NOTE: For test condition A6, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

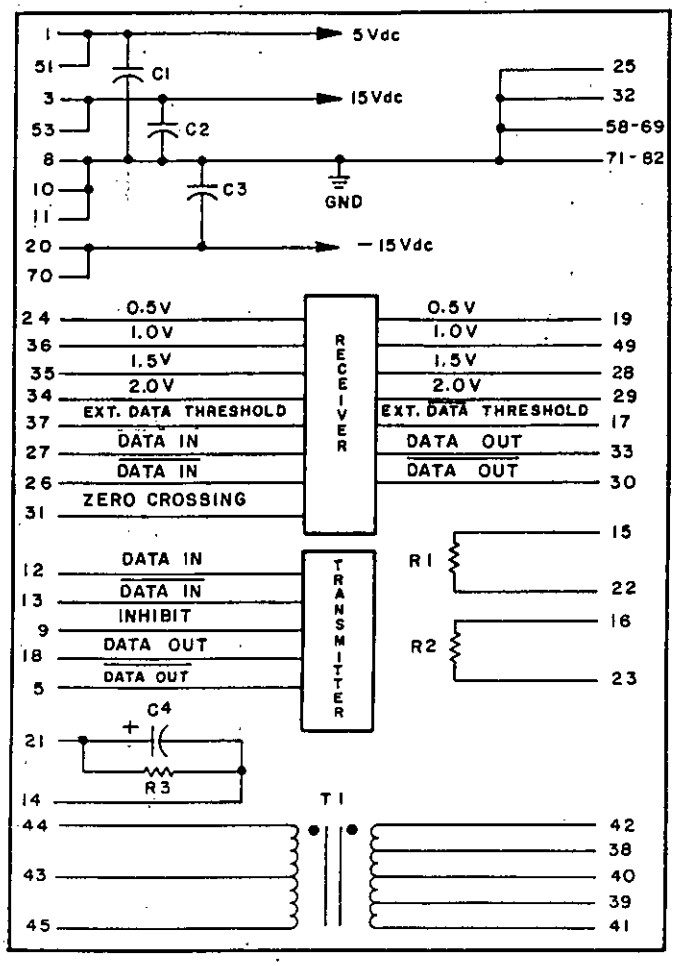


FIGURE 4. Isolation test circuit for table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Continuity	1	51	0	1	Ω	5
B2		3	53				
B3		20	70				
B4		8	11				
B5			25				
B6			32				
B7			58				
B8			59				
B9			60				
B10			61				
B11			62				
B12			63				
B13			64				
B14			65				
B15			66				
B16			67				
B17			68				
B18			69				
B19			10				
B20			71				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE III. Continuity test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B21	Continuity	8	72	0	1	Ω	5
B22			73				
B23			74				
B24			75				
B25			76				
B26			77				
B27			78				
B28			79				
B29			80				
B30			81				
B31			82				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

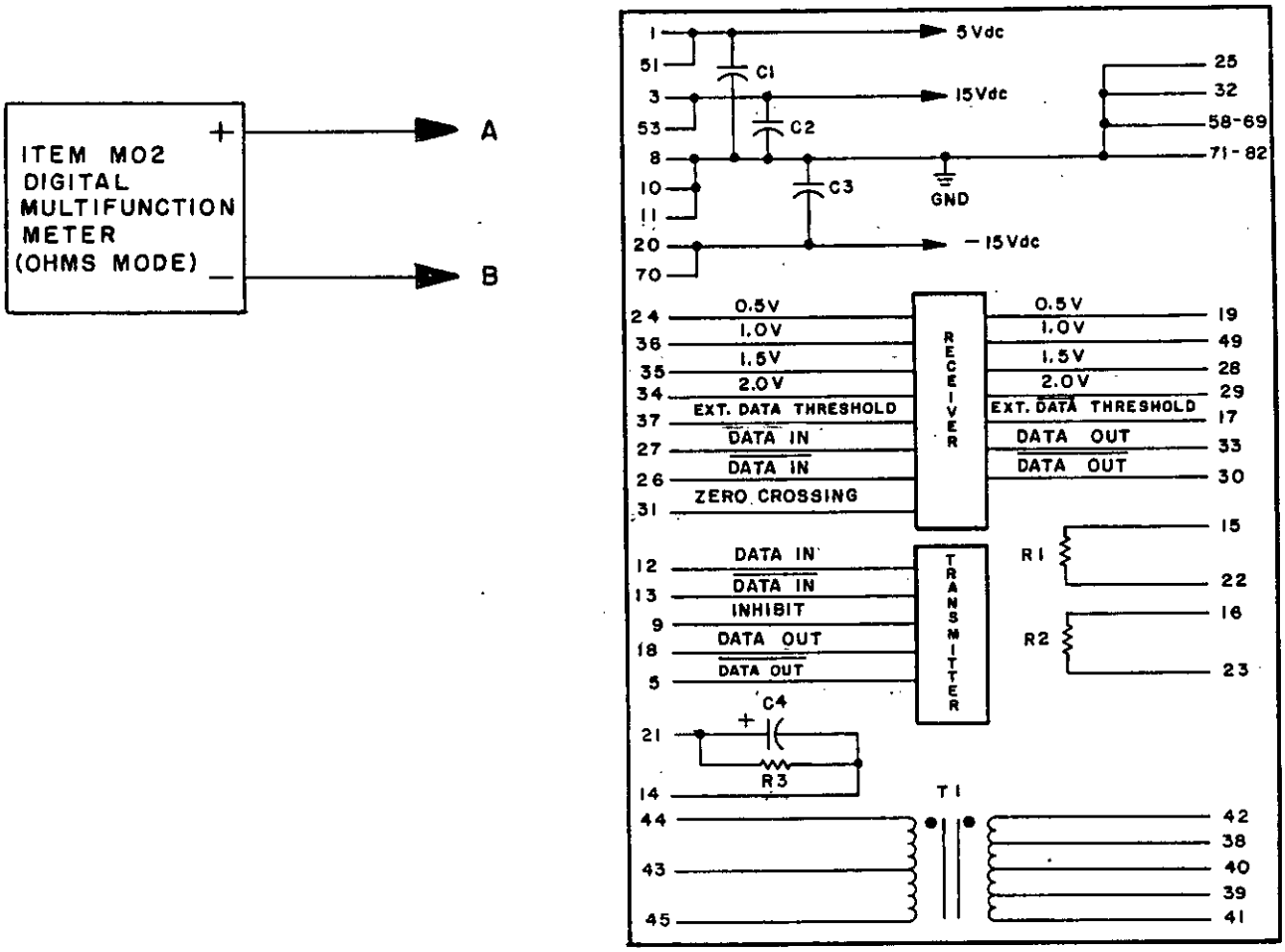


FIGURE 5. Continuity test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE IV. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
C1	Power supply current (5 V)	1	72	mA	6
C2	Power supply current (15 V)	1	95	mA	6
C3	Power supply current (-15 V)	1	95	mA	6

TABLE V. Resistance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Resistance	15	22	54.35	55.45	Ω	7
D2	Resistance	16	23	54.35	55.45	Ω	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

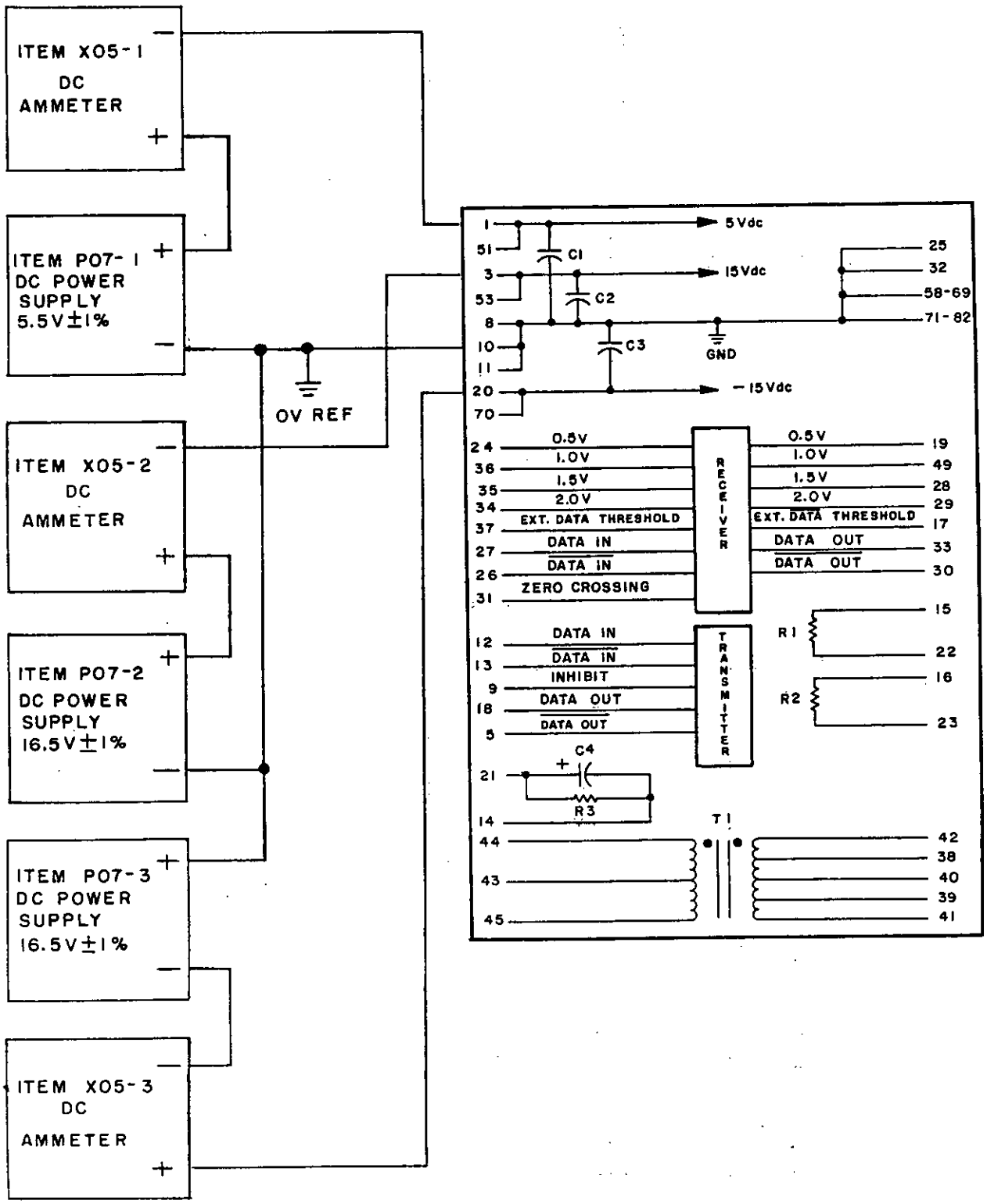


FIGURE 6. Power supply current test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

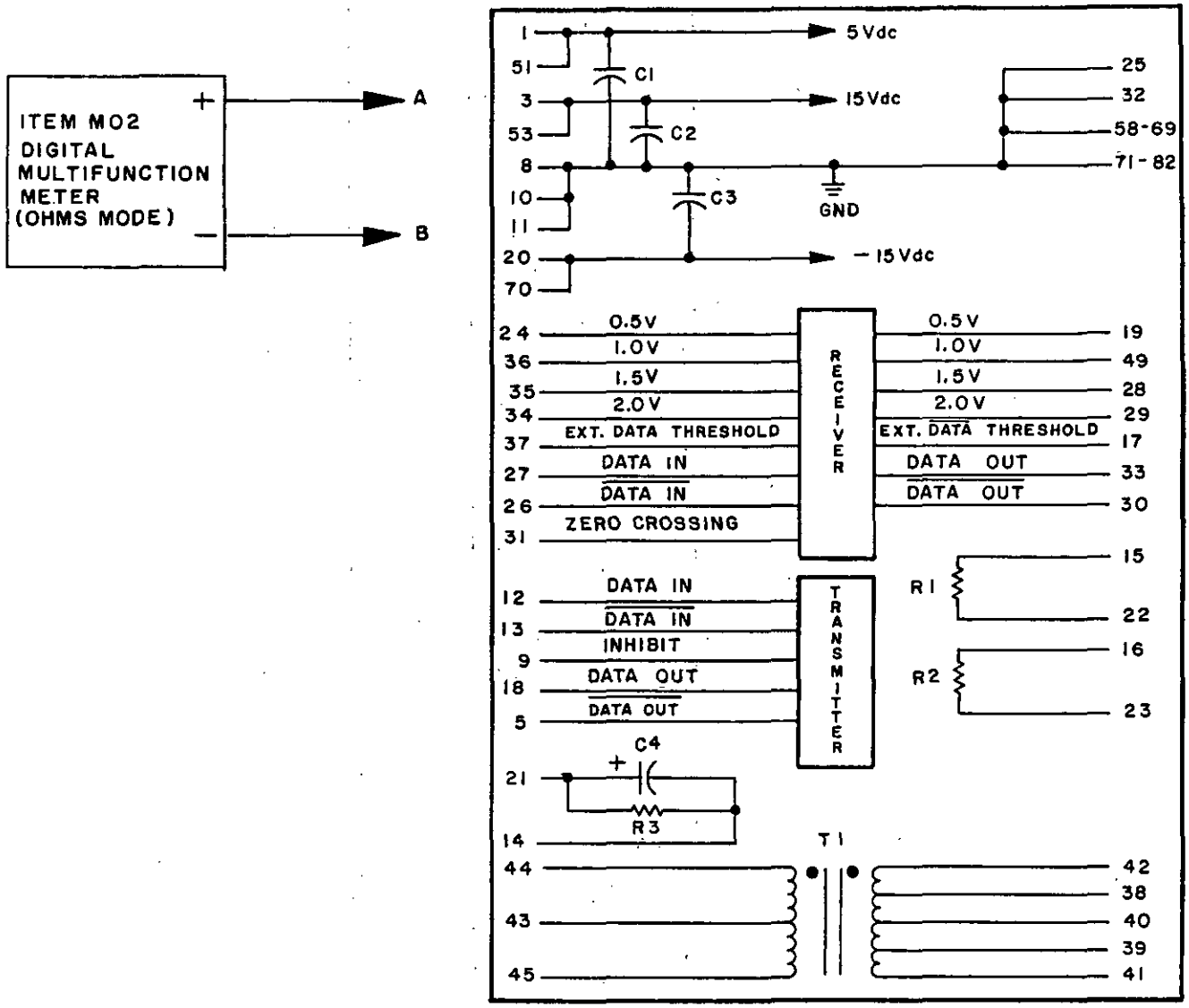


FIGURE 7. Resistance test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

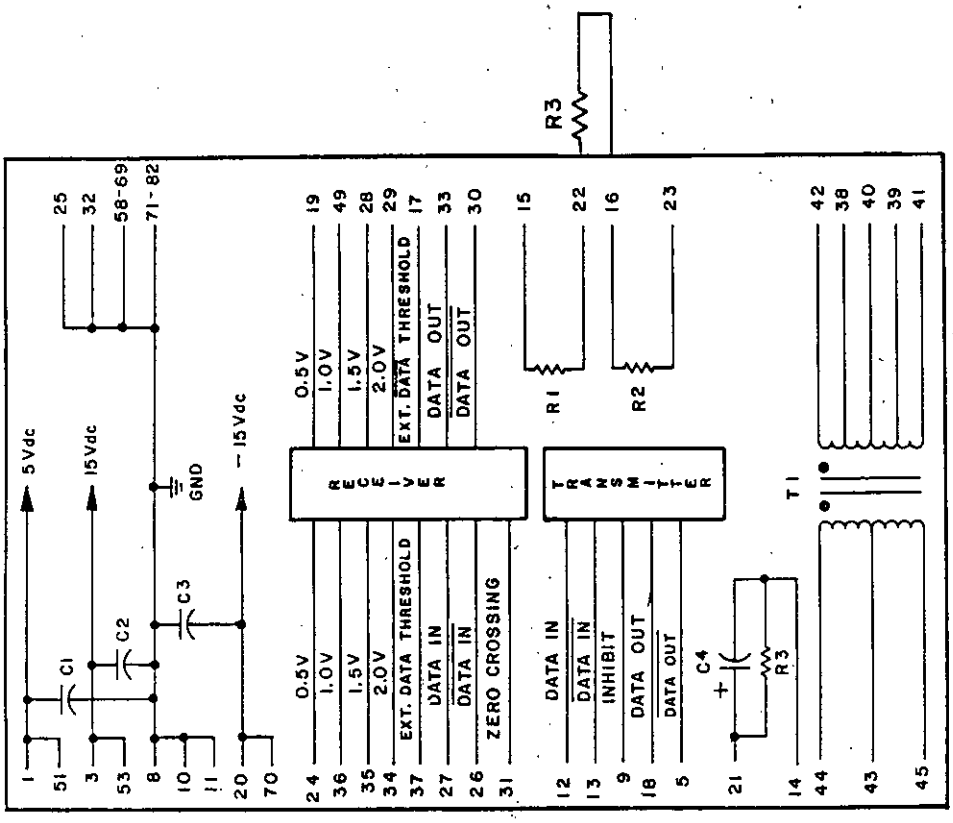
TABLE VI. Impedance (RC network) check test.

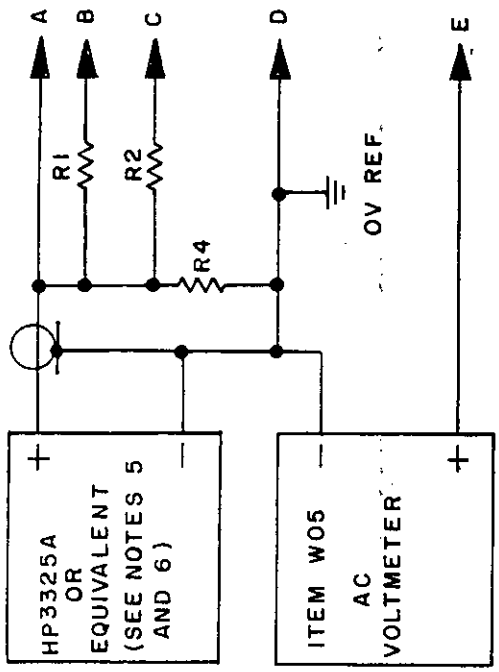
Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E	Impedance (RC network) check	Open	Open	21	14	21	2.4	3.8	V rms	8

TABLE VII. Impedance (transformer) check and output voltage (transformer) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F1	Impedance (transformer) check primary at 100 kHz	Open	44	Open	45	44	6.5	13.0	V rms	8
F2	Impedance (transformer) check secondary at 100 kHz		42		41	42				
F3	Impedance (transformer) check primary at 1 MHz		44		45	44				
F4	Impedance (transformer) check secondary at 1 MHz		42		41	42				
F5	Output voltage (transformer) at 200 kHz	44	Open		15,41,45	23,42	12.87	13.13		
F6	Output voltage (transformer) at 1 MHz	44	Open		15,41,45	23,42	12.87	13.13		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.





NOTES:
1. R1 = 5 kΩ ±1%, 1/8 W.
2. R2 = 499Ω ±1%, 1/8 W.
3. R3 = 35Ω ±2%, 1/2 W, noninductive.
4. R4 = 50Ω, 1 W.
5. Adjust HP3325A for a 500 ±5 Hz sinewave at 13 V rms ±1% for test condition E.
6. Adjust HP3325A as follows:
   100 ±1 kHz for test conditions F1 and F2.
   1 MHz ±10 kHz for test conditions F3, F4 and F6.
   200 ±2 kHz for test condition F5.
7. Test condition F1 verifies transformer primary and secondary impedances are greater than 5 kΩ.

FIGURE 8. Impedance (RC network and transformer) check and output voltage (transformer) test circuit for tables VI and VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE VIII. Receiver functional test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Receiver functional test, zero crossing voltage high level (V_Z)	16	31	29,34	2.4	5.0	V	9,10
G2	Receiver functional test, zero crossing voltage low level (V_Z)				0	800	mV	
G3	Receiver functional test, zero crossing delay time (t_{Z1})				0	300	ns	
G4	Receiver functional test, zero crossing delay time (t_{Z2})				0	300	ns	
G5	Receiver functional test, data out voltage, high level		33		2.4	5.0	V	
G6	Receiver functional test, data out voltage, low level				0	800	mV	
G7	Receiver functional test, data out response (t_{DLH}), threshold = 2.0 V					300	ns	
G8	Receiver functional test, data out response (t_{DHL}), threshold = 2.0 V							
G9	Receiver functional test, data out response (t_{DLH}), threshold = 1.5 V			28,35				
G10	Receiver functional test, data out response (t_{DHL}), threshold = 1.5 V			28,35				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE VIII. Receiver functional test - Continued.

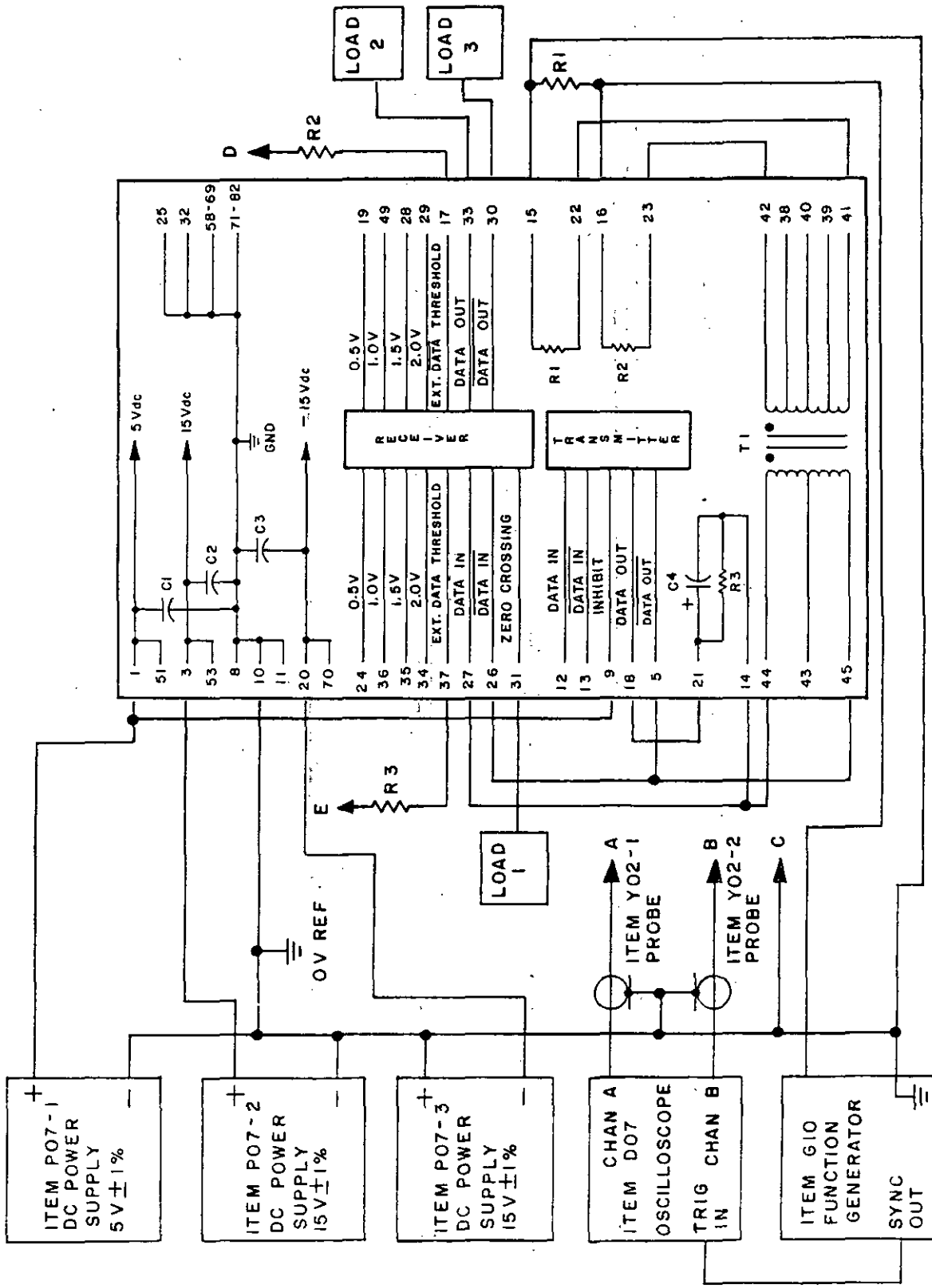
Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G11	Receiver functional test, data out response (t_{DLH}), threshold = 1.0 V	16	33	49,36	0	300	ns	9,10
G12	Receiver functional test, data out response (t_{DHL}), threshold = 1.0 V			49,36				
G13	Receiver functional test, data out response (t_{DLH}), threshold = 0.5 V			19,24				
G14	Receiver functional test, data out response (t_{DHL}), threshold = 0.5 V							
G15	Receiver functional test, $\overline{\text{data out}}$ response (t_{D-LH}), threshold = 0.5 V		30					
G16	Receiver functional test, $\overline{\text{data out}}$ response (t_{D-HL}), threshold = 0.5 V							
G17	Receiver functional test, $\overline{\text{data out}}$ response (t_{D-LH}), threshold = 1.0 V			49,36				
G18	Receiver functional test, $\overline{\text{data out}}$ response (t_{D-HL}), threshold = 1.0 V			49,36				
G19	Receiver functional test, $\overline{\text{data out}}$ response (t_{D-LH}), threshold = 1.5 V			28,35				
G20	Receiver functional test, $\overline{\text{data out}}$ response (t_{D-HL}), threshold = 1.5 V			28,35				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE VIII. Receiver functional test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G21	Receiver functional test, data out response (t_{D-LH}), threshold = 2.0 V	16	30	29,34	0	300	ns	9,10
G22	Receiver functional test, data out response (t_{D-HL}), threshold = 2.0 V			29,34				
G23	Receiver functional test, external threshold (t_{D-LH}), threshold = 2.0 V			D,E				
G24	Receiver functional test, external threshold (t_{D-HL}), threshold = 2.0 V			D,E				
G25	Receiver functional test, data out voltage, high level			29,34	2.4	5.0	V	
G26	Receiver functional test, data out voltage, low level			29,34	0	800	mV	

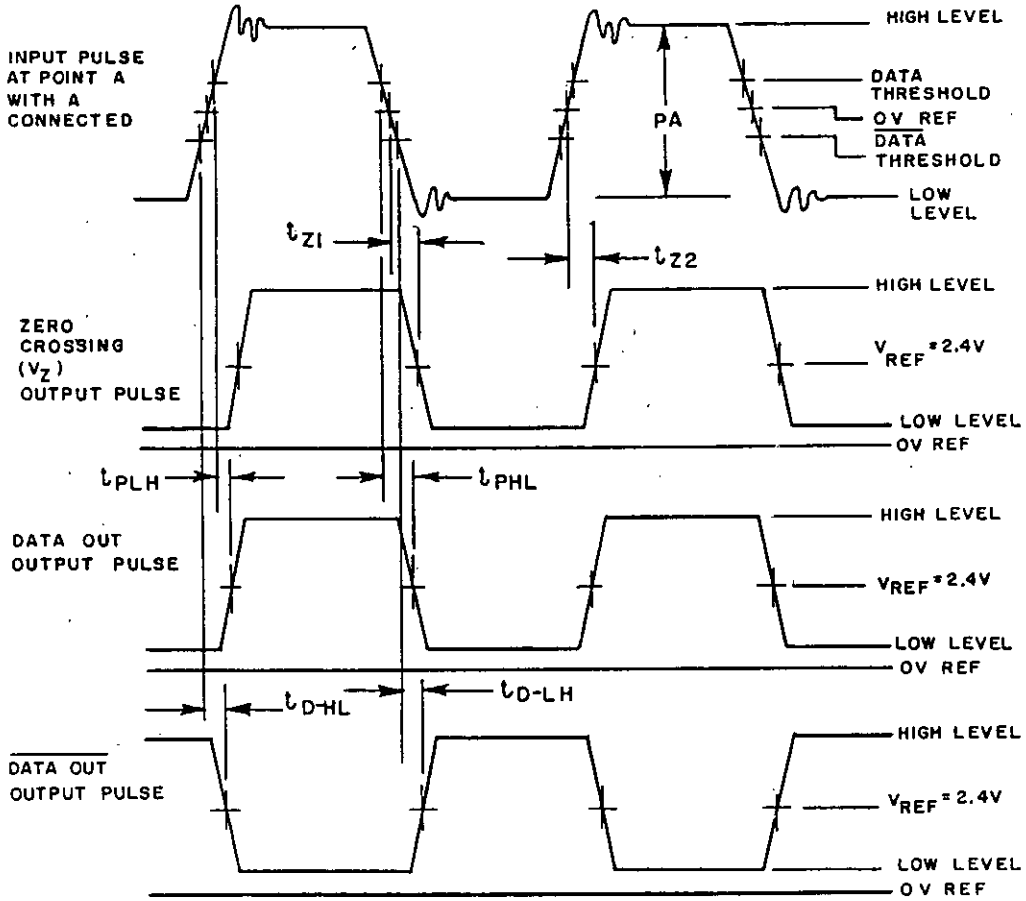
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 35Ω ±2%, 1/2 W, noninductive.
2. R2 = R3 = 10 kΩ ±1%, 1/8 W.
3. Load circuits 1, 2, and 3 are as shown on figure 11.

FIGURE 9. Receiver functional test circuit for table VIII.

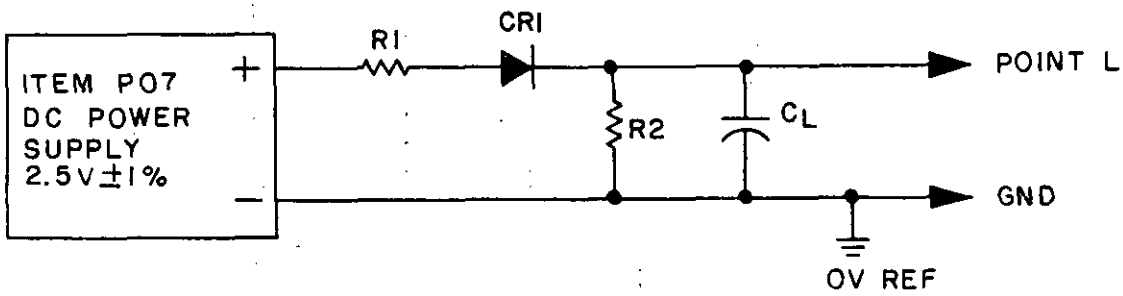
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
PA = 7 V_{p-p} ±1% for test conditions G1-G12 and G17-G26.
= 1.2 V_{p-p} ±1% for test conditions G13-G16.
PRR = 1.0 MHz ±10 kHz.

FIGURE 10. Receiver functional test circuit waveforms for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N4150 or equivalent.
2. C_L = 50 pF ±20%.
3. R1 = Select value for a current flow of 16 mA ±5% out of point L with point L held at 0.4 V.
4. R2 = 6 kΩ ±1%, 1/8 W.
5. Lead length from load circuit to module output pin under test is to be less than 1/2 inch (12.7 mm).

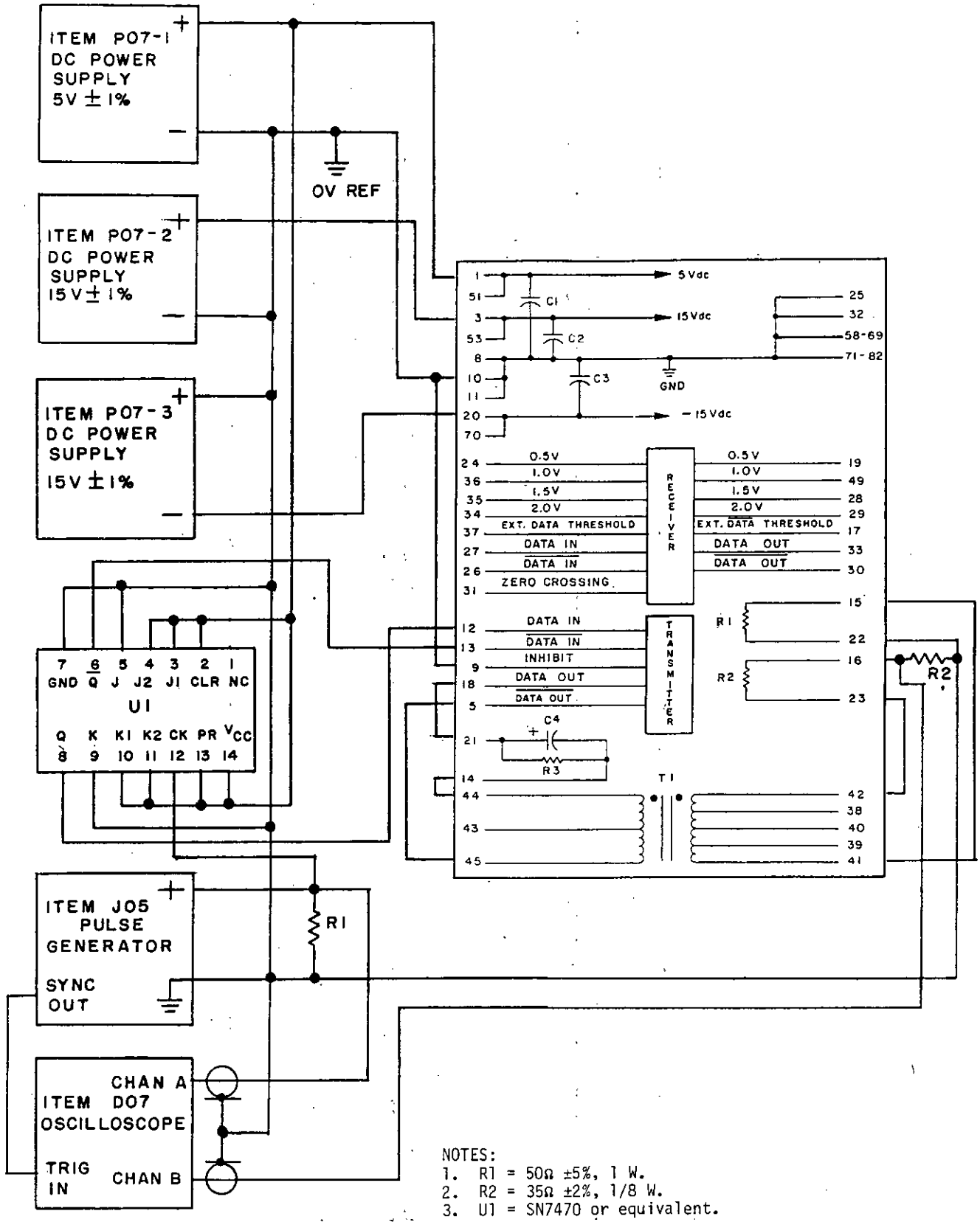
FIGURE 11. PHP logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE IX. Transmitter functional test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
H1	Transmitter functional test, zero crossing time (t_Z)	0.9	1.1	µs	12,13
H2	Transmitter functional test, transition time (t_{THL})	100	300	ns	
H3	Transmitter functional test, transition time (t_{TLH})	100	300		
H4	Transmitter functional test, propagation delay time (t_{pHL})	0	750		
H5	Transmitter functional test, propagation delay time (t_{pLH})		750		
H6	Transmitter functional test, overshoot (V_{OS})		300	mV	
H7	Transmitter functional test, undershoot (V_{US})		300	mV	
H8	Transmitter functional test, output voltage (high level)	3.0	4.5	V pk	
H9	Transmitter functional test, output voltage (low level)	-3.0	-4.5	V pk	

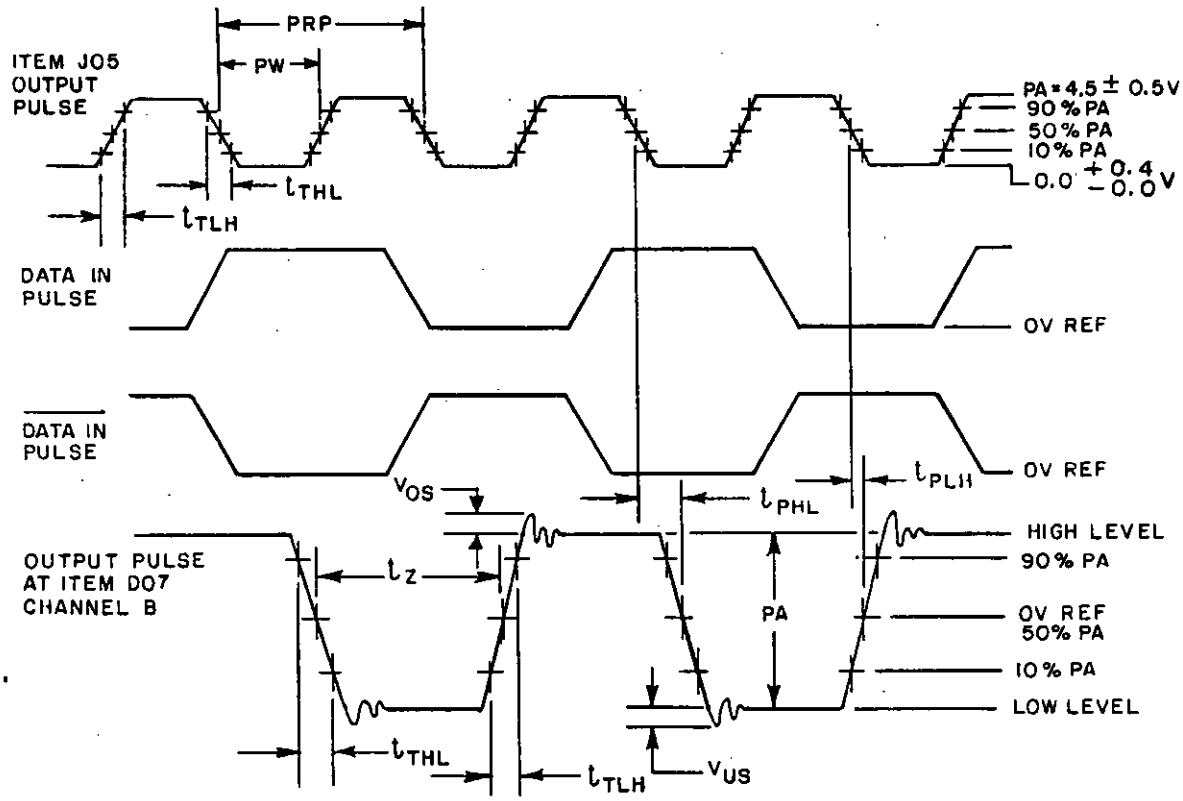
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 35Ω ±2%, 1/8 W.
3. U1 = SN7470 or equivalent.

FIGURE 12. Transmitter functional test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PW = 625 ±60 ns.
   PRP = 1.25 ±0.12 µs (800 kHz).
   $t_{TLH} = t_{THL} = \leq 20$ ns.
2. Data in and $\overline{\text{data in}}$ waveforms are shown for reference.

FIGURE 13. Transmitter functional test circuit waveforms for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE X. Receiver common mode test.

Test condition	Parameter	Connect		Initial limits 25 ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Receiver common mode, zero crossing voltage (V_Z), high level	31	29,34	2.4	5.0	V	14,15
J2	Receiver common mode, zero crossing voltage (V_Z), low level			0	800	mV	
J3	Receiver common mode, zero crossing delay time (t_{Z1})			0	300	ns	
J4	Receiver common mode, zero crossing delay time (t_{Z2})			0	300	ns	
J5	Receiver common mode, data out voltage, high level	33		2.4	5.0	V	
J6	Receiver common mode, data out voltage, low level			0	800	mV	
J7	Receiver common mode, data out response time (t_{DLH}), threshold = 2.0 V				300	ns	
J8	Receiver common mode, data out response time (t_{DHL}), threshold = 2.0 V						
J9	Receiver common mode, $\overline{\text{data out}}$ response time (t_{D-HL}), threshold = 2.0 V	30					
J10	Receiver common mode, $\overline{\text{data out}}$ response time (t_{D-LH}), threshold = 2.0 V	30					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

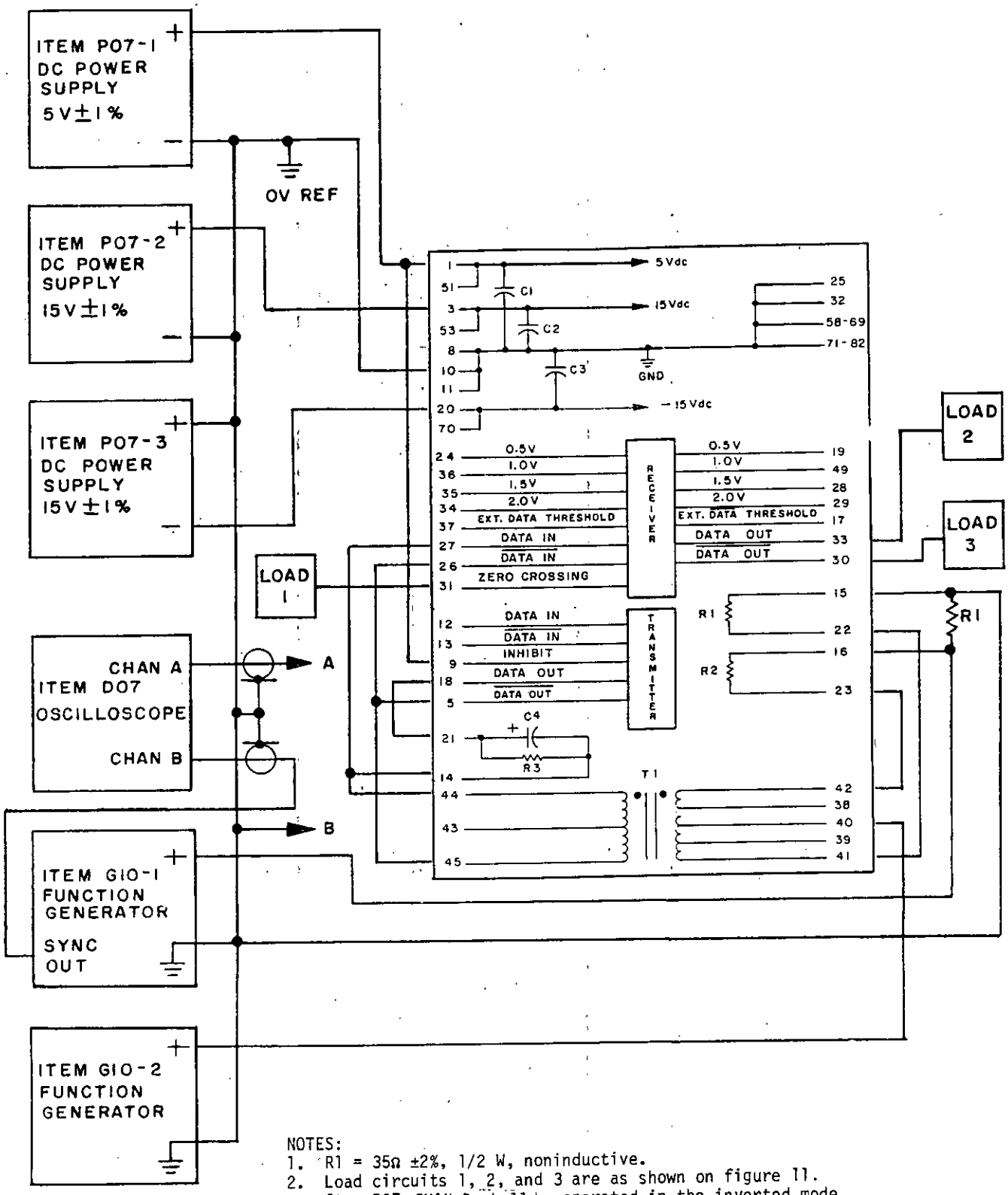
TABLE X. Receiver common mode test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J11	Receiver common mode, data out voltage, high level	30	29,34	2.4	5.0	V	14,15
J12	Receiver common mode, data out voltage, low level	30	29,34	0	800	mV	14,15

TABLE XI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Power supply filter capacitance (15 V)	3	11	5.4	8.2	µF	16
K2	Power supply filter capacitance (-15 V)	58	70	5.4	8.2	µF	16
K3	Power supply filter capacitance (5 V)	1	11	12	18	µF	16

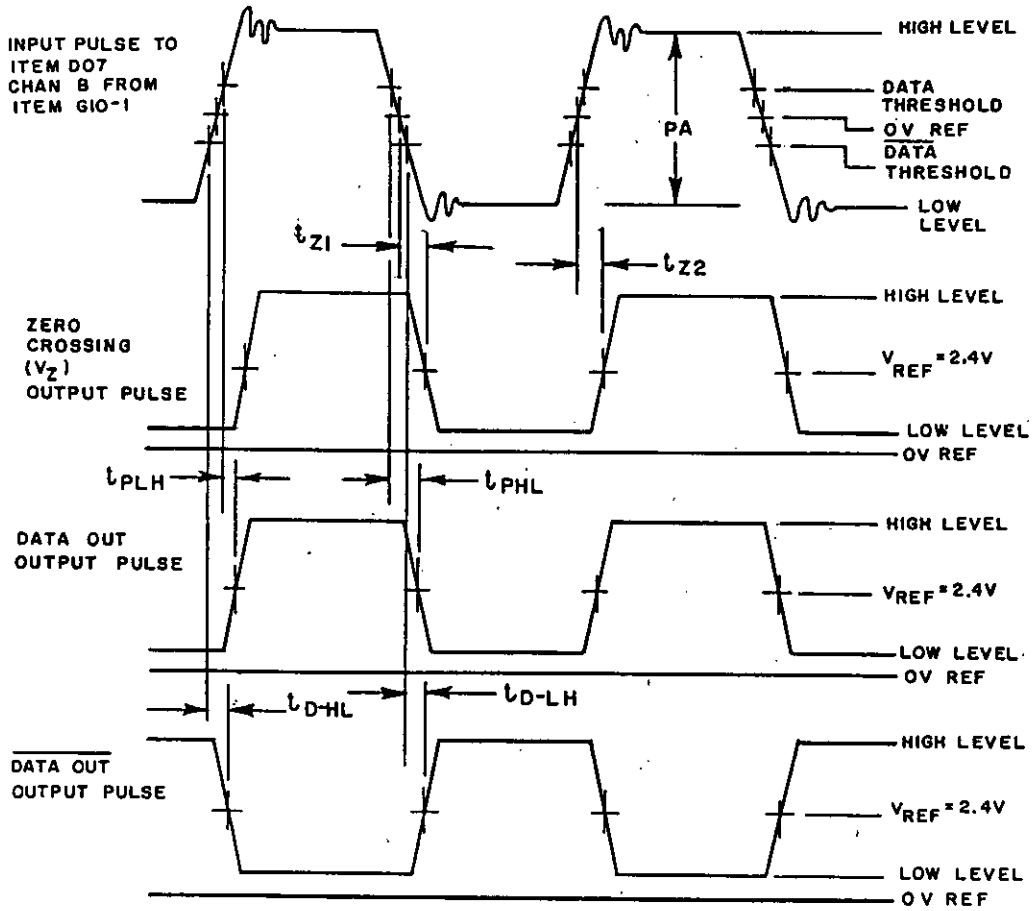
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 35Ω ±2%, 1/2 W, noninductive.
2. Load circuits 1, 2, and 3 are as shown on figure 11.
3. Item D07, CHAN B shall be operated in the inverted mode.
4. Modules which meet the requirements of table X, meet the common mode requirements of this specification.

FIGURE 14. Receiver common mode test circuit for table X.

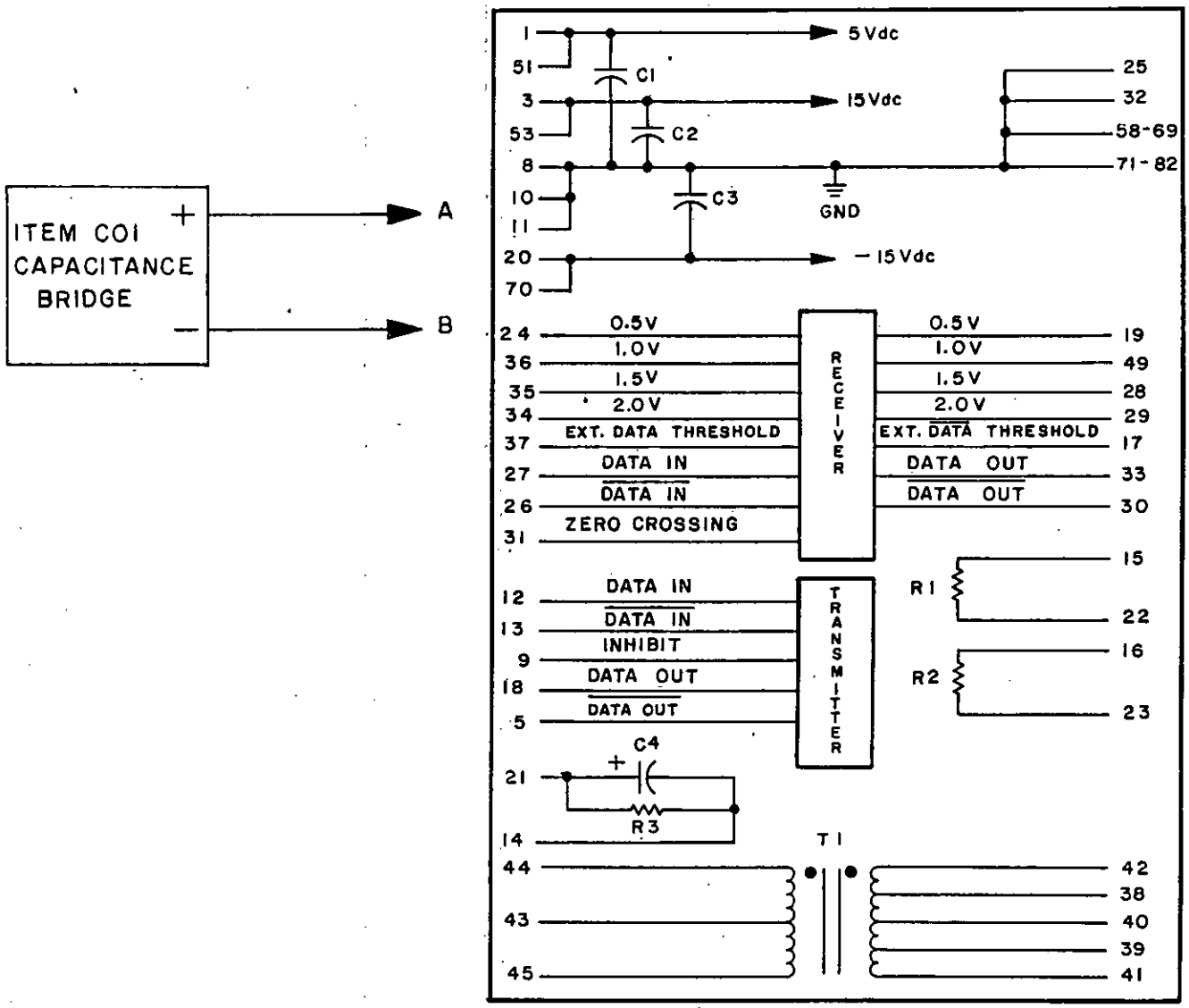
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:
Item G10-1:
PA = 7.0 V_{p-p} ±1%.
PRR = 1.0 MHz ±1%.
Item G10-2:
PA = 5.0 V rms ±1%.
PRR = 1.0 MHz ±10%.

FIGURE 15. Receiver common mode test circuit waveforms for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 1 kHz ±10%.
   V_{pk} = 0.3 V maximum.

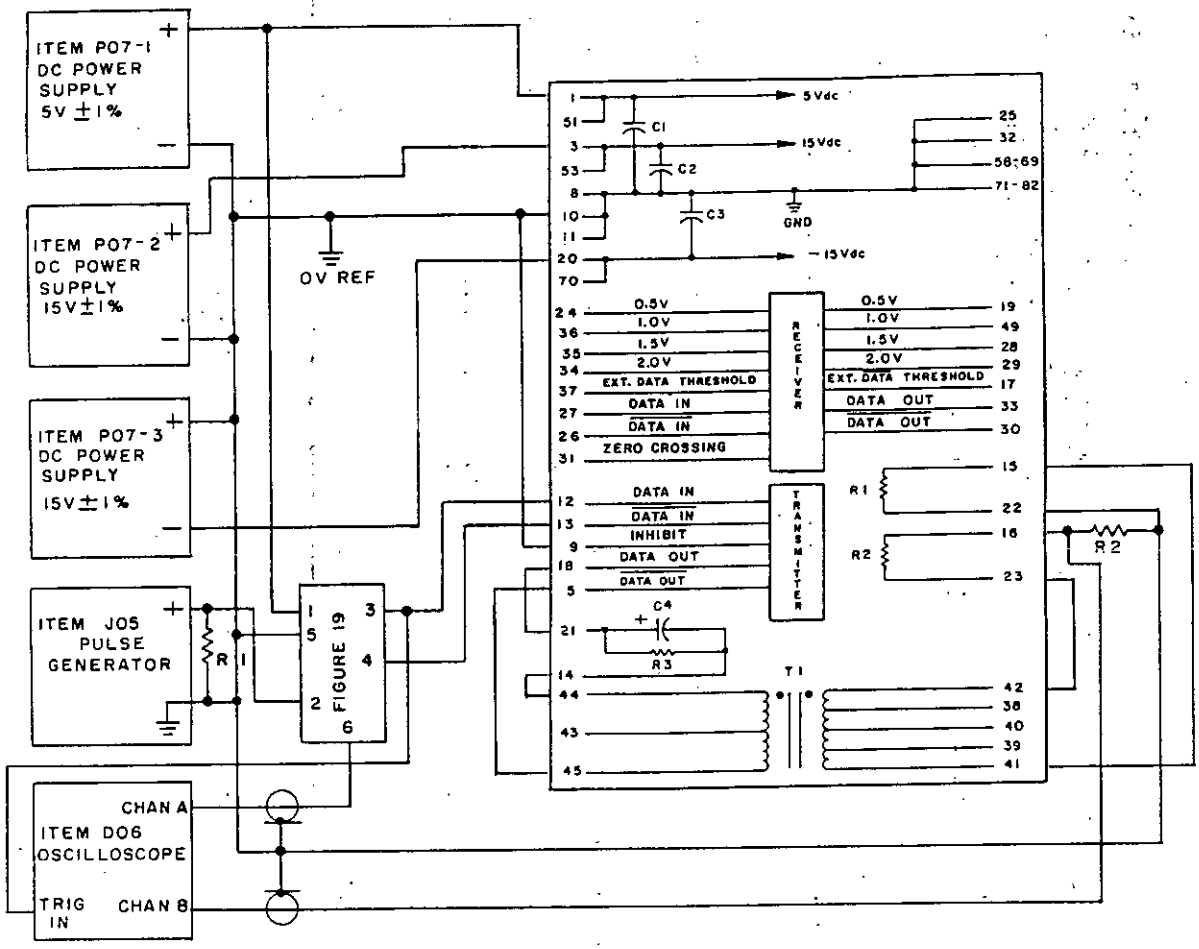
FIGURE 16. Power supply filter capacitance test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

TABLE XII. Output symmetry test.

Test condition	Parameter		Switch number																Initial limits 25 ±5°C ambient		Unit	Test figure
			S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	Min	Max		
L1	V_p	Output symmetry	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	0	90	mV pk	17,18
L2	V_n		1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2				
L3	V_p		2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1				
L4	V_n		2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1				
L5	V_p		2	1	1	1	2	1	1	1	2	1	1	1	2	1	1	1				
L6	V_n		2	1	1	1	2	1	1	1	2	1	1	1	2	1	1	1				
L7	V_p		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2				
L8	V_n		2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2				
L9	V_p		2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1				
L10	V_n		2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1				
L11	V_p		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2				
L12	V_n		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2 = 35Ω ±2%, 1/2 W, noninductive.
3. Figure 19 is a schematic of a Manchestor word generator providing data words only.
4. Set item D06 to single sweep and reset the trigger.
5. Load the word generator as per figure 19 to start each test.

FIGURE 17. Output symmetry test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

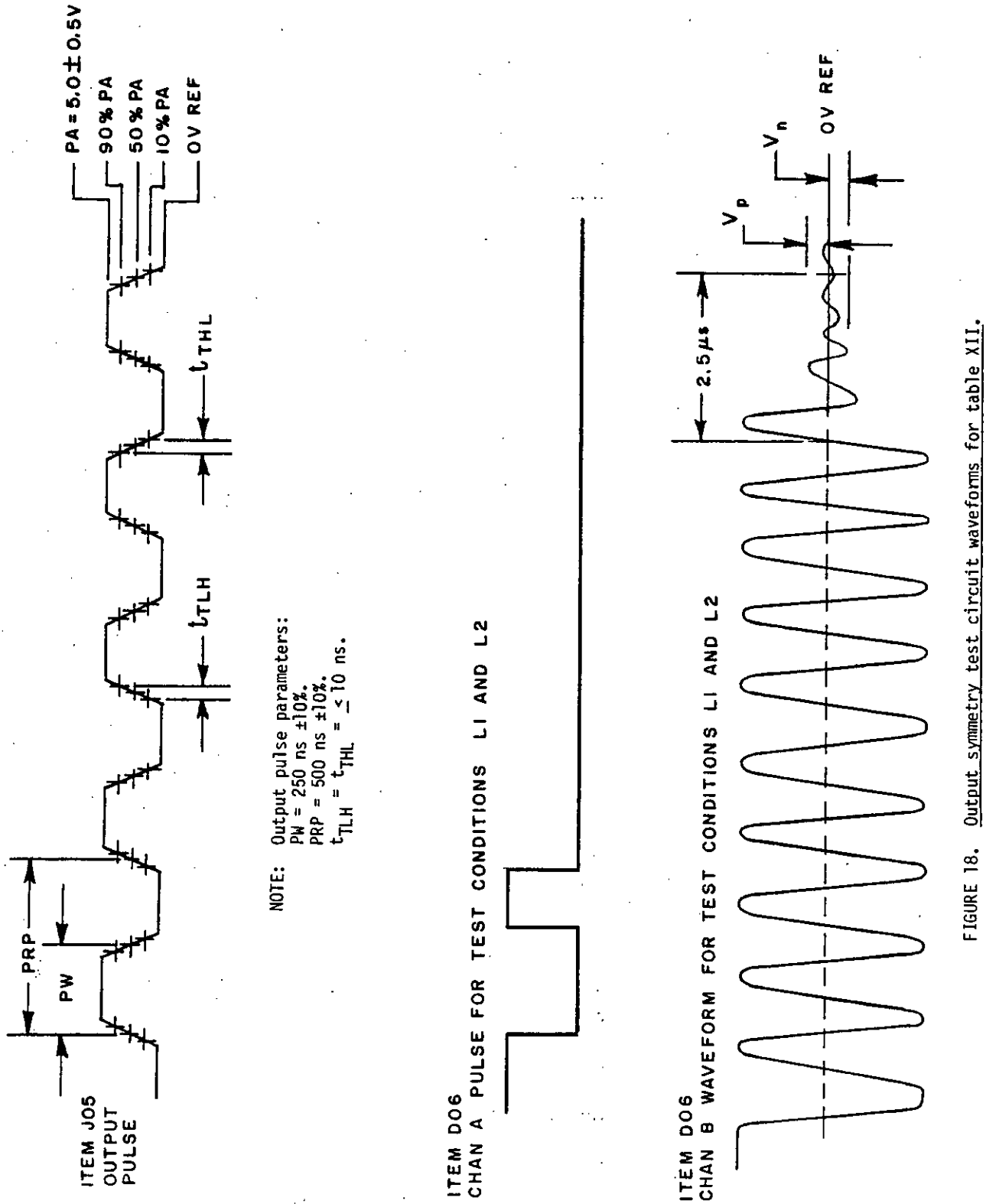


FIGURE 18. Output symmetry test circuit waveforms for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

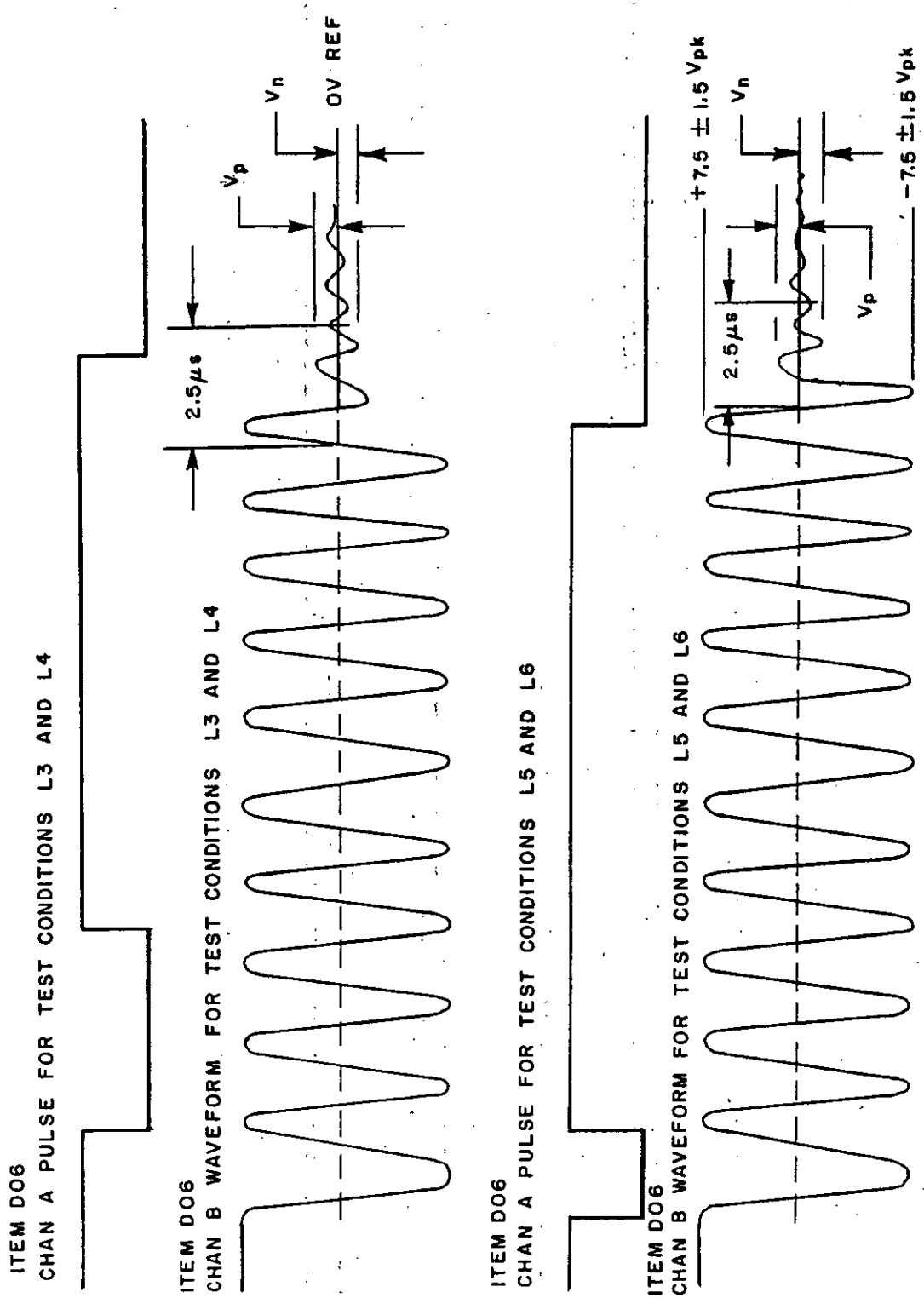


FIGURE 18. Output symmetry test circuit waveforms for table XII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

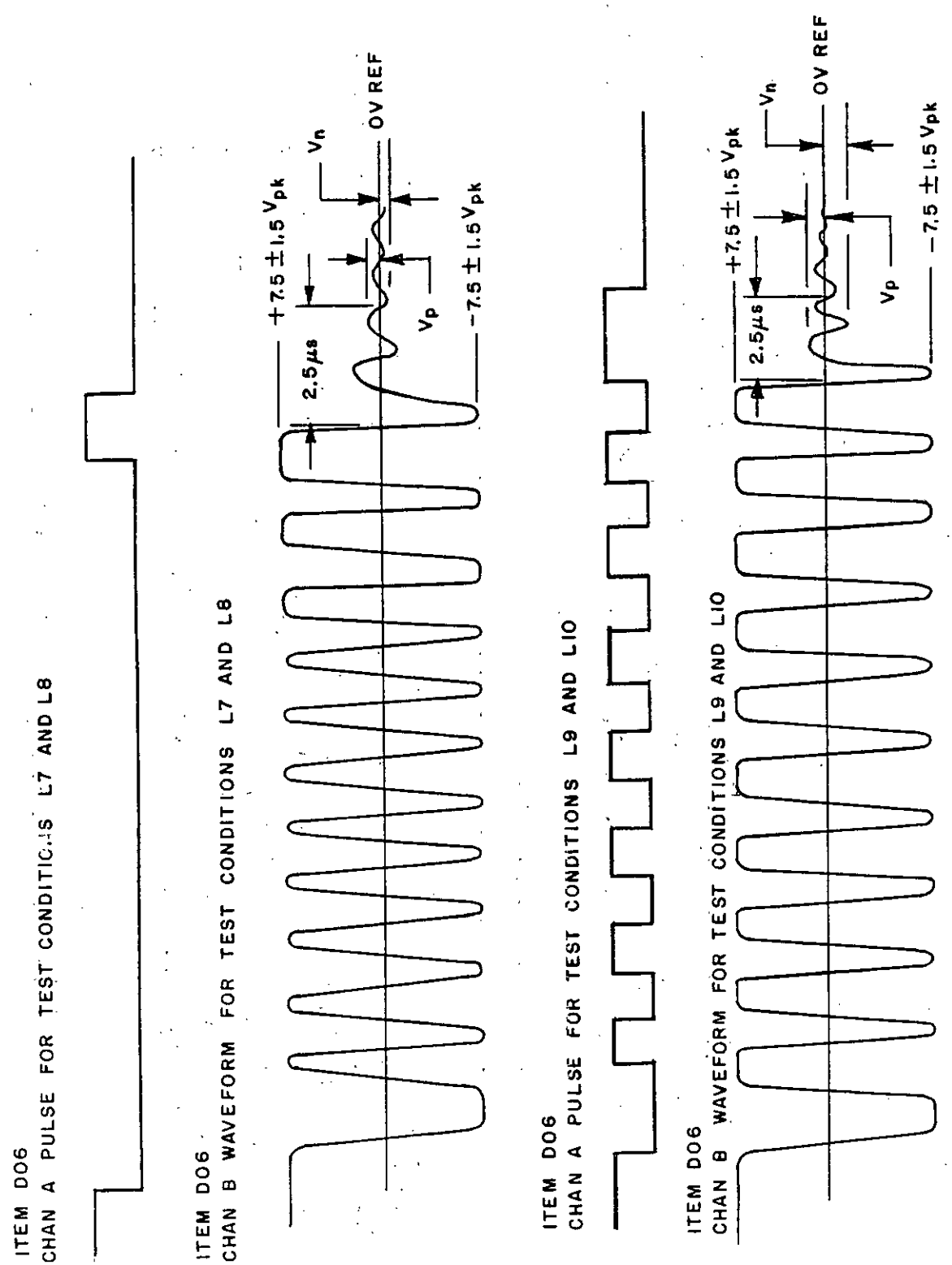


FIGURE 18. Output symmetry test circuit waveforms for table XII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

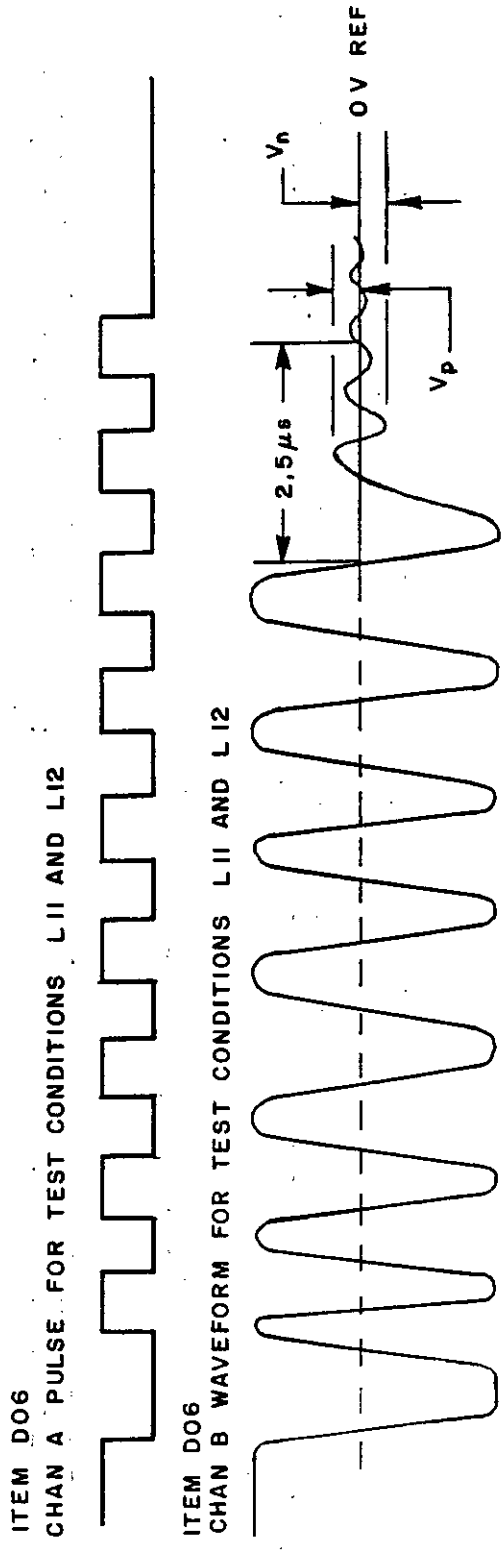


FIGURE 18. Output symmetry test circuit waveforms for table XII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

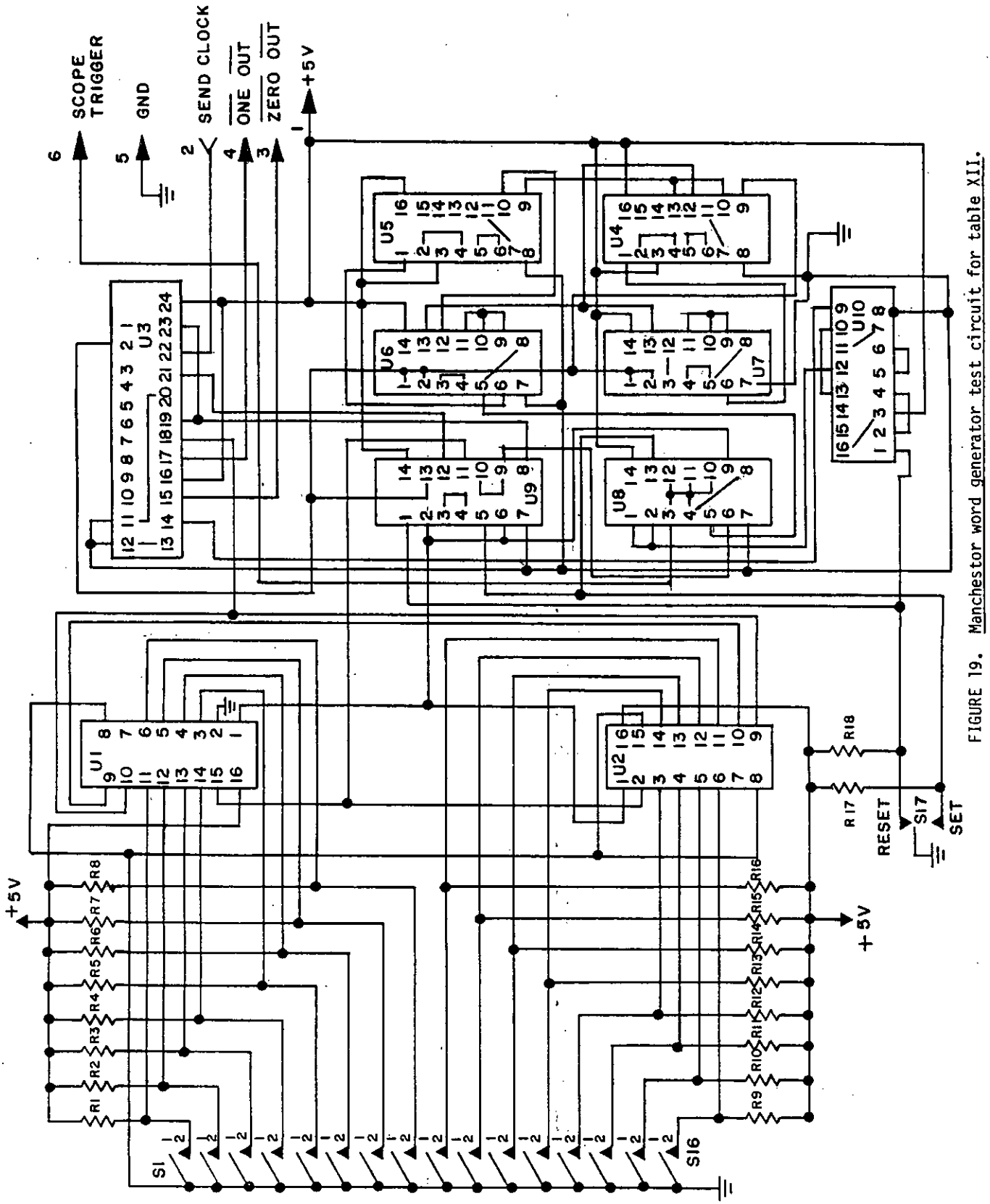


FIGURE 19. Manchestor word generator test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

NOTES:
1. R1-R18 = 1 kΩ ±5%, 1/8 W.
2. S1-S16 = SPST switch.
3. S17 = SPDT switch.
4. U1 = U2 = SN54165 or equivalent.
5. U3 = HD15530 or equivalent.
6. U4 = U5 = U10 = SN54174 or equivalent.
7. U6-U9 = SN5400 or equivalent.
8. Set switches S1-S16 as shown in table XII.
9. Set switch S17 to "reset" and then return to "set". The word generator is now loaded and ready.
10. The circuit will automatically trigger item D06 and disable the word generator after thirty repetitions of the word being transmitted. Item D06 will trigger at the last enable pulse.

FIGURE 19. Manchestor word generator test circuit for table XII - Continued.

☆U.S. GOVERNMENT PRINTING OFFICE: 1982-605-034/5649

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, D. C. 20363

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20363

ATTN: ELEX 8111

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/341(EC); Modules, Standard Electronic Interface Bus, Key Code PHP

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:54Z
Check the source to verify that this is the current version before use.